

Leveraging Ubiquitous Computing for Empathetic Routing: A Naturalistic Data-driven Approach

Arash Tavakoli
at5cf@virginia.edu
University of Virginia
Charlottesville, Virginia, USA

Mehdi Boukhechba
mob3f@virginia.edu
University of Virginia
Charlottesville, Virginia, USA

Arsalan Heydarian
ah6rx@virginia.edu
University of Virginia
Charlottesville, Virginia, USA

ABSTRACT

Although we extensively use routing services in our daily commutes, such systems are yet to be personalized around the user. It often happens that different routes are close in their estimated time of arrival (ETA) while being very different in how they affect the driver's states. Using traces of a user's physiological measures, different candidate routes can be ranked based on how they affect users' well-being. In this research, we introduce the "empathetic routing" framework for providing human-centered routing based on historical biomarkers of the drivers collected through naturalistic settings and by using smart wearable devices. Through this framework, we rank three specific routes between two points in the city of Charlottesville, based on historical driver heart rate data collected through a three-month naturalistic driving study. Additionally, we demonstrate that the proposed framework is capable of finding infrastructural elements in a route that can potentially affect a driver's well-being.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

emotion, routing, autonomous vehicles, user-centered

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1 INTRODUCTION

With the increase in automated driving assisting systems, the vehicles' in-cabin space is gradually transforming to a mobile built environment in which the driver and passengers spend less time focusing on driving and more time on other preferred and personal activities (e.g., enjoy the outdoor scenery, hold meetings, etc. [15]).

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In contrast to other static built environments (e.g., buildings), the outdoor conditions in vehicles is not static. The changes in scenery, traffic patterns, noise levels, road types may significantly impact the drivers' states, comfort, and behaviors. For instance, research suggests road features (e.g., roadside vegetation [3, 21]), road type [22], traffic conditions (e.g., traffic density [1]), and driver's familiarity with the route [13] can positively and negatively impact driver's emotional states and behaviors. This suggests route options could significantly influence the driver and passengers emotional states prior to reaching their destination.

While routing selection services have improved in providing accurate arrival time with real-time updates, traffic conditions, or avoidance options (e.g., toll roads, highways, etc.), the majority of the route optimization services provide options only based on identifying the fastest route or shortest distance to the final destination. In some cases the routing service provide options for choosing highway or avoiding toll road. However, the effect of the environment on the driver can go well beyond the sole presence of a highway when contextualizing the routes and identifying the best route based on the each drivers' context preferences. For instance, studies have showed that some people prefer passing through freeways, some prefer roads with more turns, and some avoid turning as much as possible [10]. As a result, route selection services should choose the optimum routes that not only minimize the trip duration but also provide the best driving experience to the level of improving drivers' well-being and taking their preferences into account.

For route selections to not only minimize the trip duration but also provide the optimum driving experience that may also improve the driver's state, personalized models are needed to understand individual's preferences and how different environmental/infrastructural elements (e.g., intersections, presence of a cyclist) might impact driver's states and behaviors. With the current advancements in ubiquitous computing, many drivers are already using wearable devices to track their health and daily physical activities. By having access to their real-time and historical physiological data collected through these wearable devices, we can now couple the changes in driver's states and the outdoor contextual information (e.g., traffic density, road types, number of intersections, etc.). By fusing the user's physiological data with the corresponding changes in the outdoor environment, we can learn how different infrastructural elements may impact drivers' physiological, behavioral, and emotional states. Ultimately, this information can be used to recommend routes that include elements that best fit the preference profiles of each individual driver.

In this paper, we introduce the *empathetic routing* framework for humanizing routing services through ubiquitous computing

techniques. We provide a preliminary study on using driver’s physiological factors such as heart rate (HR) and arm movement (IMU sensor) as well as the corresponding environmental attributes such as noise level, number of intersections, road geometry, and traffic density to showcase how an individual’s psychophysiological states might change given different route options. Specifically, using longitudinal naturalistic data, we showcase ranking three different routes between two pre-specified points in the city of Charlottesville based on their “empathetic score”. We further present a case study in which we have categorized different neighborhoods based on drivers’ physiological responses. Overall, the aim of this research is to pave the way for building human-centered empathetic routing systems that can provide route options according to individual preferences and current emotional states.

2 BACKGROUND

In addition to the extensive literature on routing selections based on shortest distance and estimated time arrival (ETA), there is a body of literature on personalizing routes based on factors such as (1) route safety (e.g., less crime intensive routes) [20], (2) driver preferences (e.g., user’s previously chosen routes) [20], and (3) route affect (e.g., presence of more scenery)[18]. These studies highlight the importance of user-centered route selection and propose different approaches to move beyond simple ETA and provide user-specific route options.

From the safety perspective, [20] used crowdsourced crime data in the Washington DC metropolitan area to provide a safety score for different route options. The authors optimize the routing problem by considering three different functions of the safest route, shortest route, and best route, which considers both safety and length of the route. Similarly, [8] used crime rates in cities of Chicago and Philadelphia to move beyond the shortest path possible and include safety levels in route options. Using a bi-objective shortest path approach, the authors included both ETA and safety into their model to provide and rank different route options within these cities. In a more recent study, [5] provides a framework for providing safe routing that takes advantage of both crowdsourced data and official criminal statistics with an emphasis on avoiding congestion while providing safe routing.

To consider individual user-preferences in routing options, multiple studies utilized users’ historical routing choices. For instance, [6] used users’ historic GPS traces to provide personalized routes. In their study, the authors provide a methodology based on applying the coordinate descent function for learning routing parameters. In their study, authors evaluated their method on GPS traces of 85 volunteers from the state of Washington, each monitored for 57 days. Their method significantly improved the routing quality (i.e., the level that a generated route based on learnt routing parameters is similar to historical GPS traces) to fit the individual differences. Similarly, [12] used participants’ historical data points to provide route options that pass through areas the individual had previously checked in before (e.g., restaurants). Through their proposed approach, the user familiarity with the route is maximized based on segmenting the route options while keeping the number of segments to the lowest possible. In their study, authors used road network data from the city of Los Angeles for evaluating

their method in providing familiar routes. By performing an experiment on 10 participants, the authors found that more than half of their test cases preferred personalized routing provided by their method. In a recent study, [10] used context and user-generated GPS traces to provide routing in a personalized contextualized fashion. In their study, the proposed method was designed to learn a context-preference vector from the user-generated traces of GPS. The context preference vector attempts to connect the user’s preferences to different contextual elements. The context can include travel duration, time of travel, departure time, weather condition, etc. Based on the mapping of context to GPS and historical routing, they provide routing for context vectors that historical routing is not available.

From the lens of route selection based on affective states, [18] used participants responses on pictures of two street scenes in London to rank paths based on beauty. By collecting more than 3000 responses from participants, the authors identified that by adding just a couple of minutes of additional walking in a route, the user could be happier as the selected route could be quieter and more beautiful (e.g., scenic). In another study, [17] used self-reported data as well as scraping social media platforms to classify odor levels across different regions of cities of London and Barcelona and how participants ranked these areas based on perceived pleasantness. They found out that locations with more pleasant smells were associated with a more positive subjective affect. [4] introduced a framework titled *Scenic Planner*, in which they use geo-tagged images with historical users check-ins to provide more scenic route options. In their study, authors apply heuristic algorithms to maximize a defined total scenic view score for a route. A scenic score in this study is defined based on the concentration of geo-tagged images along the routes. [11] analyzed the effect of adding external factors such as users’ preferences, scenery, and affect into routing. Through using a controlled platform developed by the authors, they discovered that traveling in more scenic routing decreases the traffic density in highways and significantly increases it in locations around parks. [7] used self-reported responses of participants to detect different emotional states while commuting through different means of transportation (e.g., public transportation, personal vehicles). The authors provide a map of different areas of the city of Minneapolis, Minnesota, based on different emotions the participants provided, such as happiness and sadness.

Although recent studies indicate personalized routing options can enhance an individual’s safety, comfort, and satisfaction, the proposed approaches rely on subjective responses (self-reported data) and/or historical GPS traces to classify an individual’s preferences. In the following sections, we first present the “empathetic routing” framework and then present a case study in which we showcase how physiological biomarkers could be used to personalize routing options for different individuals.

3 PROPOSED EMPATHETIC ROUTING FRAMEWORK

Figure 1 demonstrates our proposed framework for providing a user-specific empathetic routing option by leveraging the data collected through wearable devices and ubiquitous computing techniques. As

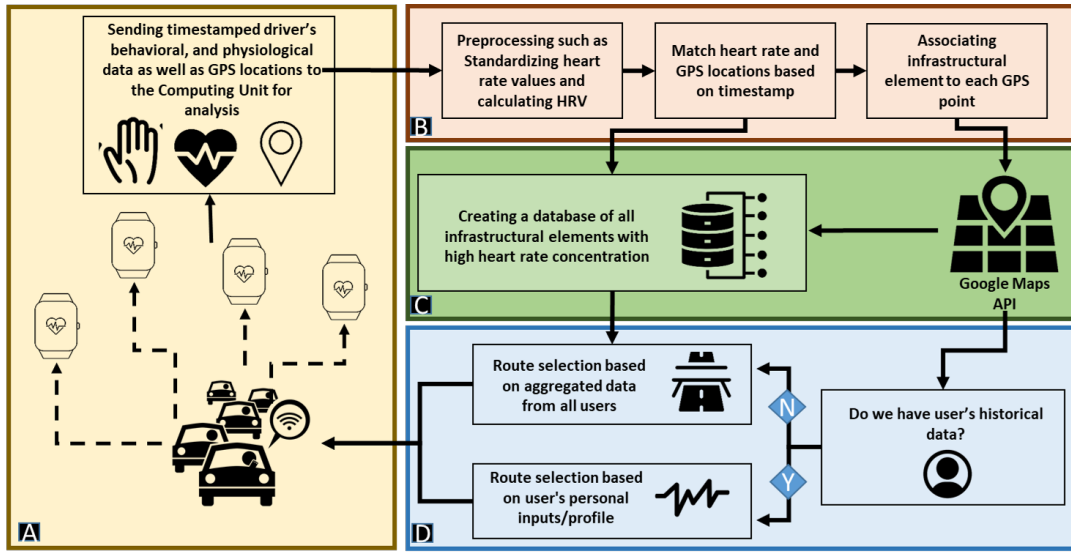


Figure 1: The empathetic routing framework

shown in this framework, we collect and process the driver’s physiological data as well as behavioral data through wearable devices to identify the changes in the driver’s state and behaviors within different routes. These physiological markers are timestamped with the GPS data allowing us to accurately identify the time and location of where individuals have a sudden change in their physiological responses (i.e., HR). The data from wearable devices is collected through a developed in-house app, namely SWear [2], that is installed on participants’ mobile devices. SWear records the driver’s HR, photoplethysmography (PPG) signals, and hand movement, along with other ambient conditions, such as light, noise, and GPS locations (1 - A).

After retrieving data from the wearable devices, we pre-process the collected data by removing low-quality data, calculating heart rate variability (HRV) features from raw PPG sensors and standardizing HR values per participant to account for individual differences (1 - B). Prior research has shown that higher HR and lower HRV values correlated with higher/lower stress levels [24] or aggressive driving [14]. As a result, in this study, we use the higher and lower levels of HR to identify timestamps and geo-locations of where participants might feel more or less stressed. Once the GPS coordinates are extracted for the specific timestamps, we use the Google Map API [9] to extract infrastructural and environmental elements around each GPS points, including the presence of certain roads, intersections, and high traffic density. Once this information is collected, a summary of the environmental elements associated with higher and lower HR values is recorded to a database (1 - C). This database includes each specific infrastructural elements, the average HR value around the infrastructural elements, and their GPS coordinates. This information, along with user-specific physiological data (if available), is then used to inform route selection services to provide personalized empathetic route options.

We can then use the created database to classify the possible route options based on an individual’s HR/stress profiles or the

aggregated data from the general population (if available). For instance, once a user requests to get to a destination, at first, all possible route options are identified through a route selection application, such as Google Map. If HR data is available for that route, our framework then assesses whether we have enough data from the specific user to provide a personalized routing or not. If yes, based on the user’s historical physiological data, the framework provides the route that is least stressful for the user. One method for this assessment is to analyze the historical average HR value in each route for the user and pick the route with the historically lowest HR value (1 - D). In the case of not having enough available data from the specific user, the framework attempts to find the best routing among the candidate routes based on other users’ aggregated data. In this option, the framework identifies the route based on (1) the lowest average standardized HR for all users and (2) minimizing the presence of infrastructural elements that correlate with a higher HR level (e.g., avoiding intersections).

4 CASE STUDY

To demonstrate how to implement the proposed empathetic routing framework, we used the HARMONY dataset, which is a longitudinal fully naturalistic driving behavior study collected in the northeastern states of the United States [23]. This dataset includes multimodal data streams collected from video cameras and wearable devices from 21 drivers for a period of three to six months (on average 18-20 hours of driving data for each participant). For this paper, we only used the wearable data streams available in HARMONY dataset for driving events that took place within the city of Charlottesville, Virginia and its surrounding areas. The wearable data includes drivers’ HR (1 Hz), photoplethysmography (PPG) signals (10 Hz), ambient light level (10 Hz), noise level (every 10 minutes), driver’s movements including, accelerometer, gyroscope, and magnetometer (10 Hz), and GPS location (every 10 minutes).

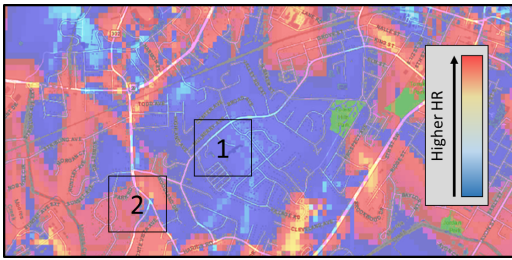


Figure 2: The heatmap of different neighbourhoods using HR values collected through driver's smart wearable devices. The more red regions are associated to higher heart rate values

At first, we standardized each participant's HR data to account for individual differences. Since each participant might have different baseline HR values, we standardized every participant's HR data based on their HR mean and standard deviation values. Then we build a heat map of drivers' HR over the city of interest. We used the triangular irregular interpolation (TIN) option in the QGIS software [16] to superimpose the HR values over different neighborhoods of the city. Figure 2 shows that different parts of the city can vary in their overall contribution to variation in the driver's HR. For instance, as it can be noted, there is a significant difference in participants' HR levels between neighbourhoods #1 and #2. This is inline with previous findings providing the effect of the different urban environment (e.g., residential vs. commercial areas in a city) on human's well-being [19].

Each HR data point also includes a timestamp, which we use to identify the precise GPS point. Although the HR data was collected at 1 Hz, we only considered the HR values for each participant when the GPS data points were also available. This was because the GPS data was collected with a lower frequency (due to battery consideration). In the newer version of our app, we retrieve GPS points through other sensor points such as participant's mobile devices. The same process has been performed for environmental noise and ambient light data. Overall, we identified 15,123 data points where we have access to driver's HR, GPS locations. We have excluded the quantitative analysis of ambient light and noise level data due to the low frequency of data collection within the city of Charlottesville, and we only focus on a qualitative comparison for noise sensors. The quantitative analysis will be addressed in our future work.

Based on the association between aggregated HR and their respective GPS points, we identify different infrastructural and environmental elements that are associated with higher HR values. For instance, in our data these elements included intersections, ramps, tight road segments, and high traffic density areas. We can then statistically identify the relationship between individual/group stress levels and the location and/or environmental elements.

Let us assume that a user attempts to travel between two arbitrary points in the city of Charlottesville as shown in Figure 3. At first, the system requests the routing options between points A and B, using a routing service such as Google Maps [9]. As a result, Google Maps provides us with three candidate options between



Figure 3: Three different routes chosen between points A and B. These routes have similar ETA (approximately 9-10 minutes)

the two points that are also fairly similar in their ETA (around 10 minutes). The system then uses the aggregate HR data in each of the routes to provide an overall score for each route based on all user's normalized HR data. For simplicity, here we define the score based on the drivers' aggregated HR, meaning that the route with lower HR values gets a higher score for keeping the driver in a calmer state. Note that we can perform the same analysis below for a specific user's data to make our route recommendation more personalized.

Figure 4 provides the overlay of HR values (the darker red is higher standardized HR, implying more stress) and traffic patterns map (yellow and red lines show higher traffic density) on the route map. Because the noise data collected in the study was sparse, we only refer to each route's noise level in a qualitative fashion (e.g., low, medium, and high). In each route, we identify infrastructural elements associated with high HR values and showcase how they are added to the infrastructural/environmental dataset. Route 1 passes mostly through a highway that has a relatively higher speed limit (65 mph), more noise, and fewer intersections. On average, drivers who passed through this route had experienced six infrastructural elements that were accompanied by high HR values. Specifically, on this route, the driver has to drive through three busy intersections (box #1, #2, and #8), two ramps (boxes #3 and #6), and two merging on/off highway events (box #3, and #5) (collected through Google Maps API). Additionally, due to the higher speed limit and less traffic density on the highway (collected through Google Maps API), the driver can keep a lower high rate in certain parts of the route (box #4). In this route, our HR data from all participants indicates that, on average, areas with higher speed limits and a lower number of intersections correlate with lower HR while ramps, merging events, and areas with higher noise levels show the adverse effects.

Route 2 passes through a residential neighborhood that has a relatively lower speed limit (35 mph), a lower noise level, and a higher number of intersections. On route 2, the driver passes through one ramp (box #1), two busy intersections (boxes #2 and #6), and one fairly tight road segment which has a higher traffic density (box #3). In this route, the HR data indicates that, on average, the lower speed limit, tight roadway design, and higher number of intersections correlate with higher HR levels in drivers, while areas with lower noise levels and residential housing correlate with lower HR levels in drivers. Route 3 passes through the main streets of the city. It

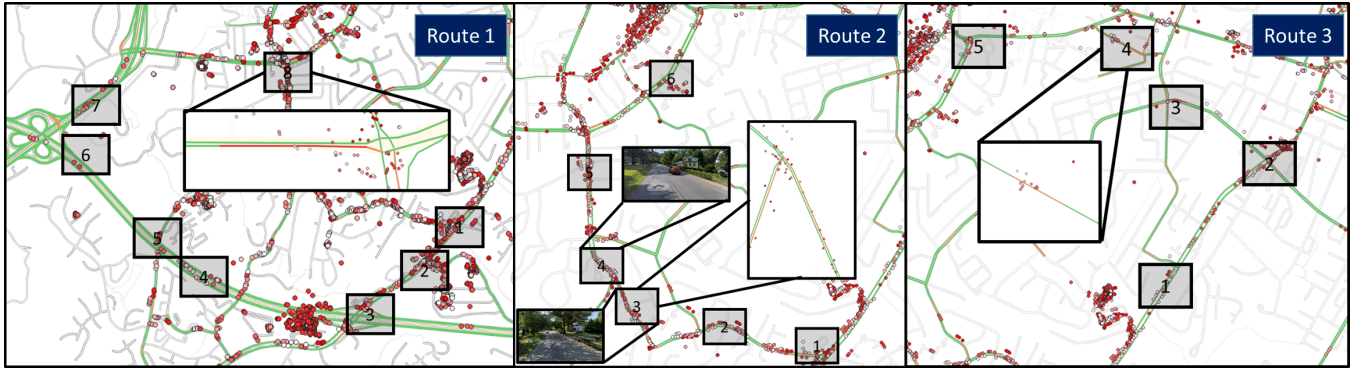


Figure 4: Analyzing the concentrated HR data points on all routes. Each numbered box is associated with one contextual element on the route such as high traffic density or presence of an intersection.

has a relatively average speed limit (45 mph), and medium noise level. On route 3, the driver passes through lower traffic density areas (box #1) and four intersections (boxes #2 to #5). The average speed limit, lower traffic density (for major parts of the road), and a lower number of intersections correlate with lower HR values among all participants.

Finally, in order to choose among the three routes, we compare the aggregated HR values on each route using Welch’s t-test [25]. We assume the collected HR values are independent, because (1) they are either collected from different people, or (2) they are selected at 10-minute intervals (due to a low frequency data collection of GPS data), which makes them far enough in time from each other. Figure 5 shows the HR values in each one of the chosen routes. The results show that all three routes are significantly different in standardized HR values (p-values of 0.0236, 0.0001, and 0.0046 for comparing routes 1 & 2, 1 & 3, and 2 & 3, respectively). Additionally, route 1 has the highest HR compared to the two other routes. Considering that route 3 has the least number of regions with high HR values, it is expected that this route is accompanied by the lowest HR values, stating a calmer driving event. Note that here we showed that by just using a simple comparison, we can assign higher scores to routes that have historically been more helpful to driver’s well-being. In the case of not having such data, we can classify route options based on the type and number of environmental/infrastructural elements that historically have shown to increase/decrease drivers’ HR levels.

5 DISCUSSION AND FUTURE WORK

In this study, we presented a framework for introducing empathetic and personalized routing selection. Compared to the previous approaches, we have used physiological data collected from 21 participants to compare three different routes in a real-world driving scenario between two arbitrary points that have very close ETA. As shown in the previous section, without any real-time or historical data from an individual or general population’s stress levels within these routes, all the identified three routes are viable driving routes. However, by integrating the HR levels, we can see one of the routes significantly performs better to keep the drivers’ stress levels lower.

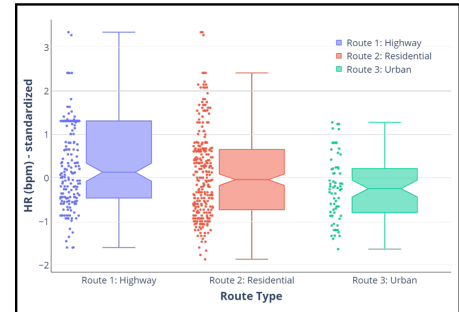


Figure 5: A comparison of historical HR values in different routes. Note that route 1 has the highest and route 3 has the lowest standardized HR values

Although in this study we have only demonstrated the application of the driver’s HR, multiple other measures provided by the smart wearable devices can also be considered. For instance, these devices collect ambient noise and light levels, as well as driver’s behavioral info such as hand movements. For light and noise levels, this information can be used for providing routes that have a more desirable ambient condition. We have not presented the results for these sensors as we require more data to statistically analyze the impact of these factors. For behavioral data, we can optimize route selections based on behavioral factors such as categorizing routes with the least number of distractions for the driver. Also, it should be noted that our smart wearable device did not collect other physiological measures such as skin temperature, and conductance. In our future work we plan to expand on the different modalities of data that can be considered in ranking/optimizing route selections.

One important implication of the presented work is to provide context vectors to different routing options that are based on an individual’s emotional and behavioral states. Such context vectors can then be calculated based on the existing methods (e.g., see [10]) to optimize routing. Note that previous studies and current routing services also consider the context by providing a choice between highways versus city or to pass through toll-free routes. However, our goal here is to move beyond simple contextual elements and to

provide an understanding of each environmental/infrastructural element on a driver's well-being. In the presented case study, we have shown that even small attributes of the road, such as tight segments of a road, can be very important when considering the driver's state.

Furthermore, the outcome of the presented framework can be used by city planners to provide a more holistic overview of the effect of different areas of the city on the drivers, or choosing different design options to potentially decrease driver's stress level. While our main focus is on driver's well-being, the proposed framework can extend to other roadway travelers, including pedestrians and cyclists. For instance, the same analyses can be performed to link pedestrians' well-being with the attributes of neighborhoods such as walkability, noise level, population density, and the amount of greenery. Note that the idea of personalizing urban environments was proposed by other researchers in the past, but bridging the gap between real-time and historical physiological sensing and infrastructural elements through our framework has not been explored before.

Our preliminary study had shortcomings mostly due to being in the pilot version and the low frequency of location data collected through our system. We are currently in the process of collecting a more diverse set of data from more participants across different cities in the United States. Additionally, in our study, we have not considered the differences between the hours of the day (morning vs. afternoon time), different genders, and driving experiences while choosing routes. These gaps will be addressed in our future work.

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